Mandibular Premolars

Presented by:

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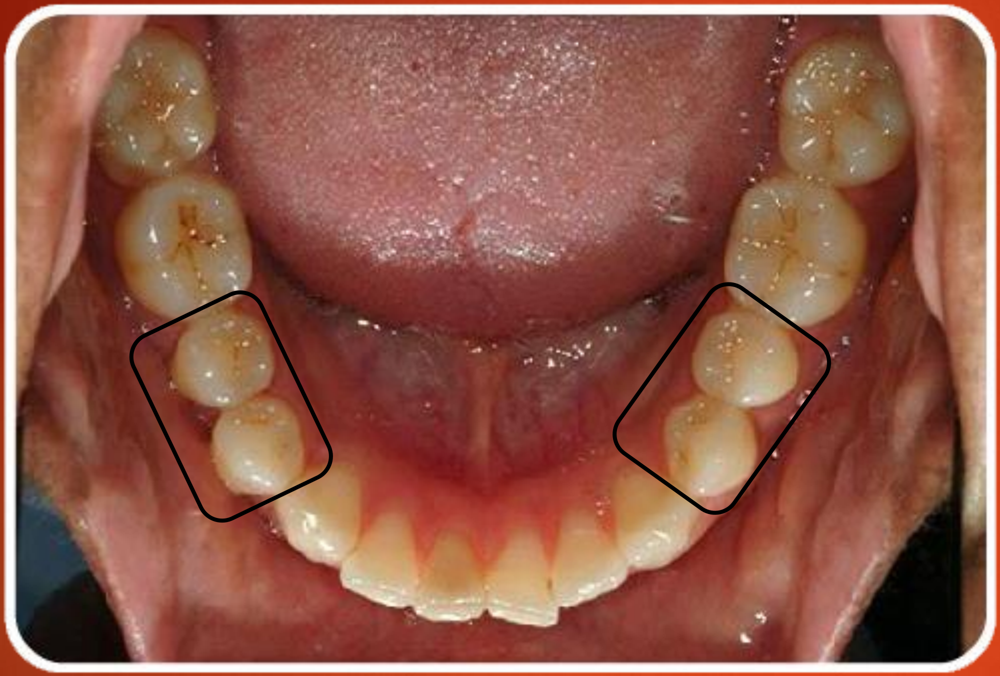
How many mandibular premolars are found in each quadrant??



The first premolar function is mainly ……......

Second premolar function in mainly…….........

Mandibular First Premolar

It has 5 surfaces:

Buccal

Lingual

Mesial

Distal

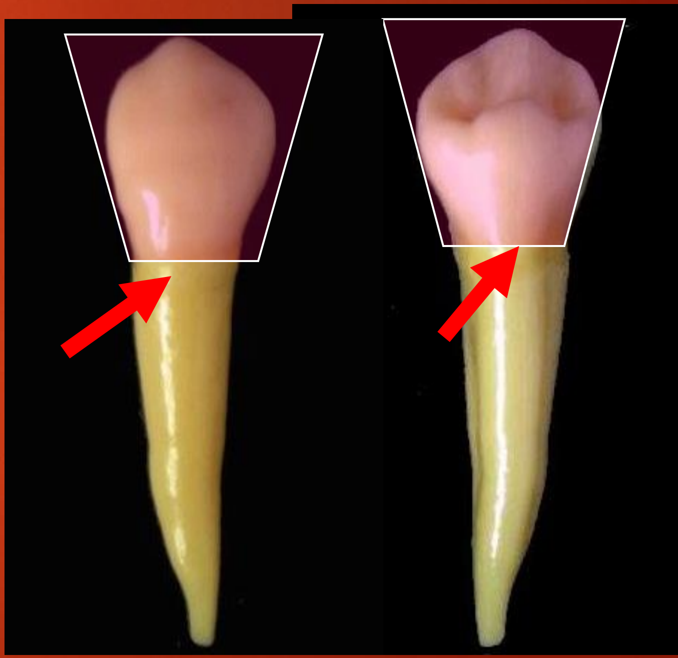
Occlusal

Buccal Surface

Geometric outline of the crown:

Buccal and lingual surface have trapezoid outline.

The smallest uneven side is cervically.



What is the Significance of trapezoid geometric outline???

Cusp tip deviated **MESIALLY**

M slope shorter than D slope and both are concave

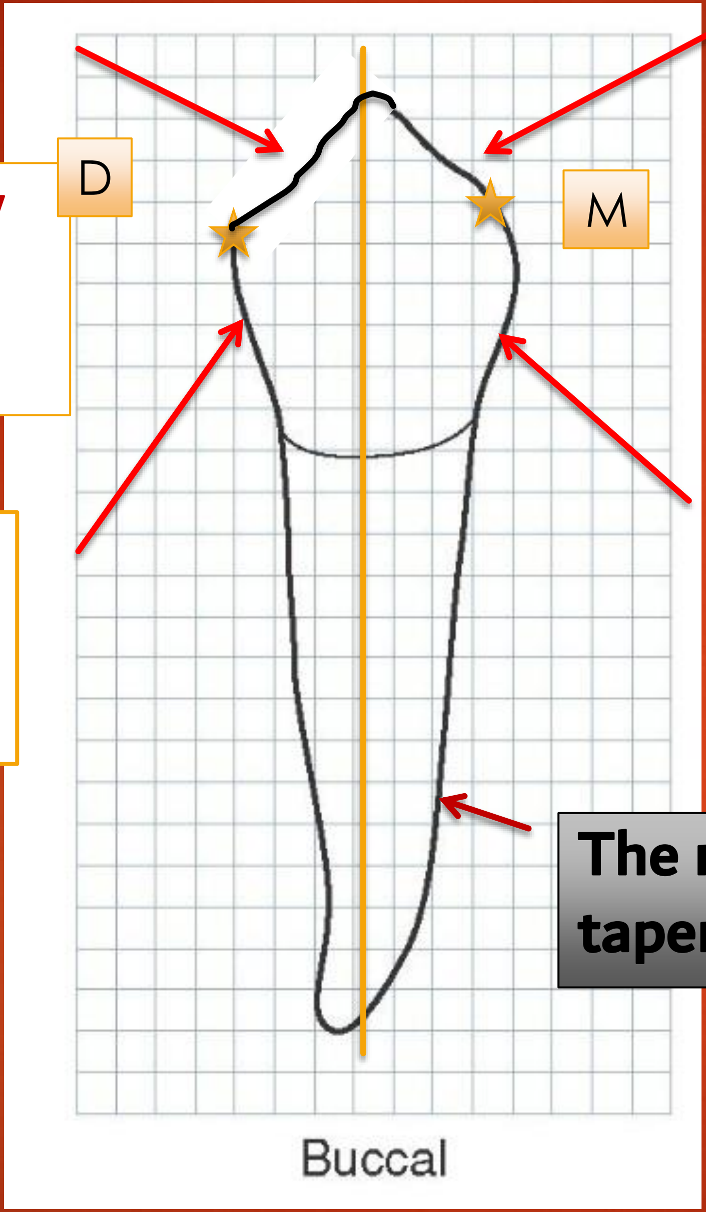
Contact distally more cervically positioned.

Contact mesially at the junction of O and M 1/3

Distal outline nearly concave

Mesial outline nearly concave

The root is cone shape with tapered distally curved apex.



Surface anatomy of the crown

Elevations:

- Cervical ridge (found in the cervical1/3).
- Buccal ridge (runs cevico-occlusally in the middle 1/3).

Depressions:

Two shallow depressions present mesial and distal to the buccal ridge.

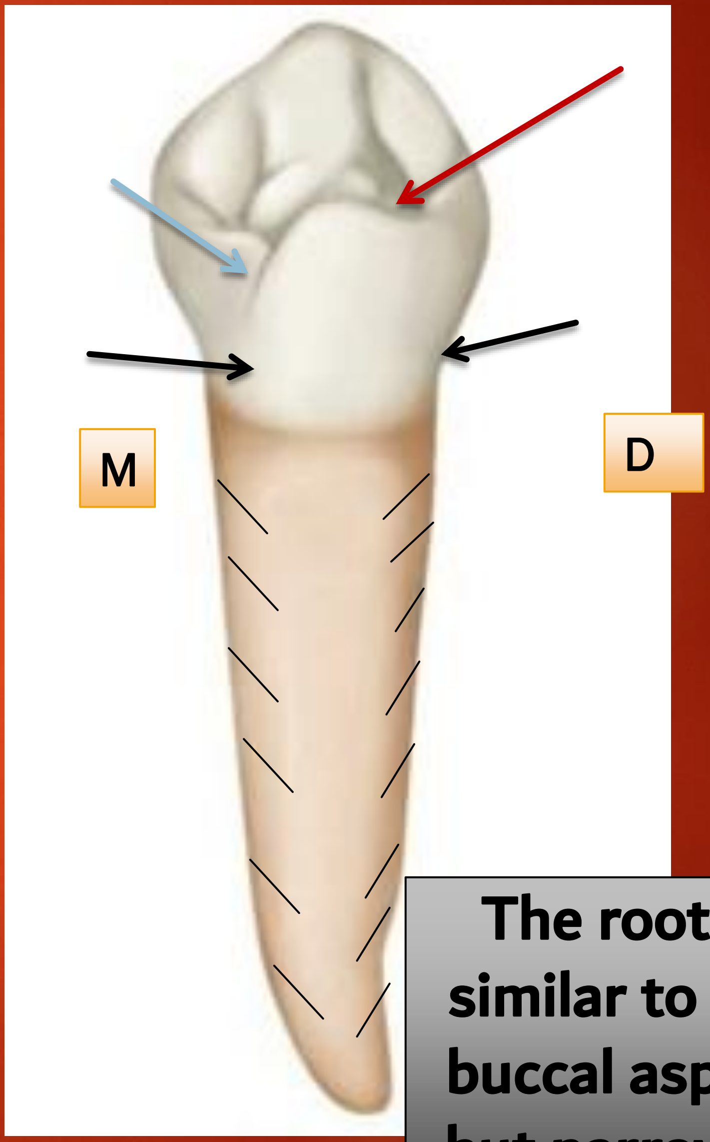
Lingual Surface

- Severe lingual convergence .

- The L cusp is short and small reaching 2/3 the crown length so most of the occlusal aspect could be seen.

- Elevations: The lingual surface is convex with the maximum convexity at middle 1/3.

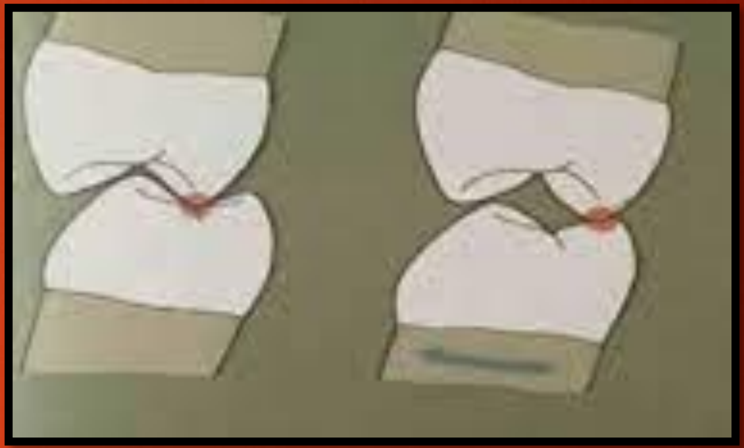
Depressions: Mesiolingual DG originates from the pit at the bottom of the mesial triangular fossa crossing the ML line angle.



The root is similar to the buccal aspect but narrower

Geometric outline of the proximal surfaces

Distal surface Mesial surface



Significance of rhomboidal geometric outline in centric occlusion

Rhomboid in shape with narrow occlusal table

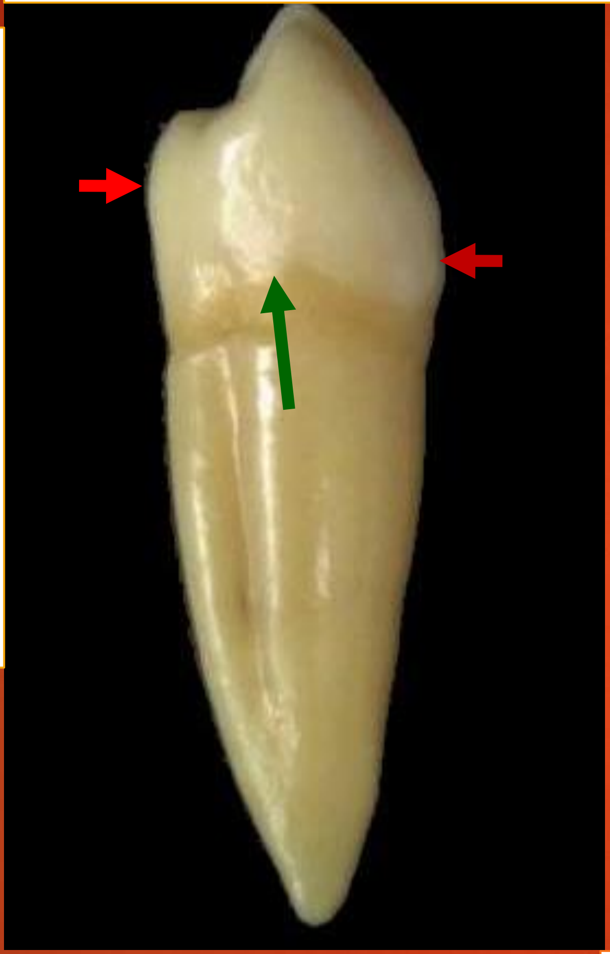
Lingual inclination is well prominent

Outlines of proximal surfaces

Mesial surface

Distal surface

Buccal outline is convex with the maximum convexity at Cervical 1/3 representing ....



Lingual outline is convex with maximum convexity at ....

Cervical line curves occlusally and more ....positioned distally

Mesial surface

Occlusal outline

The buccal cusp is centered above the root due to severe lingual inclination of the crown.

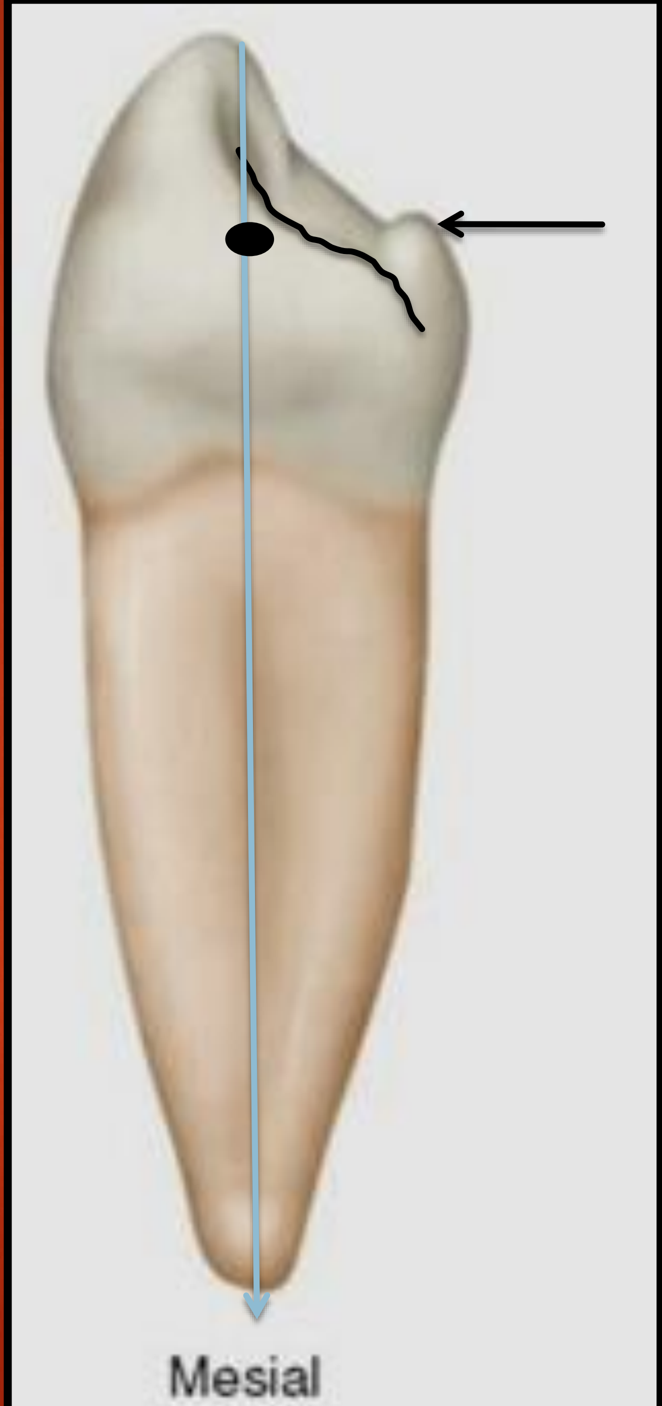
The lingual cusp is 2\3 the buccal cusp.

Surface anatomy:

MMR severely sloping bucco-lingually ending with **mesiolinguual DG.**

Mesial contact area presents on midline <u>buccolingual</u> and at .... <u>Cervico-occlusal.</u>

The root is wide and has **deep groove** with tapered end.



Distal Surface

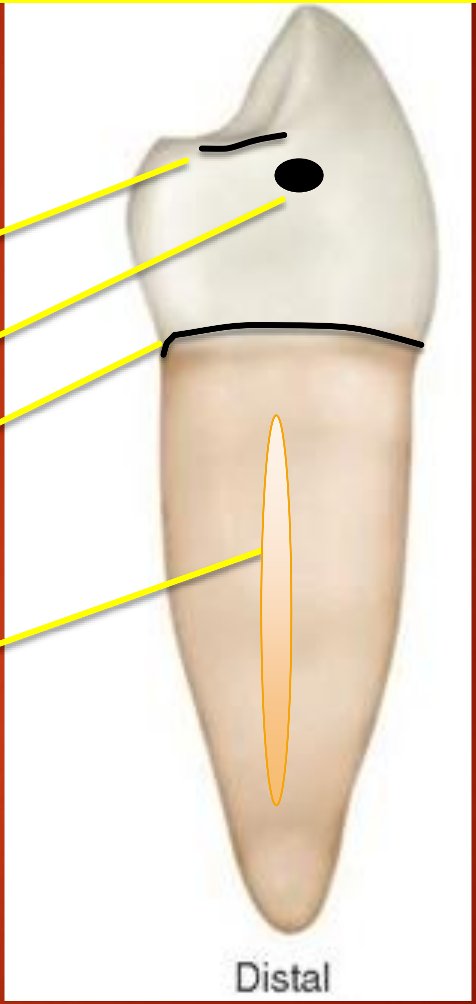
Differences with mesial surface:

1-DMR is perpendicular to long axis of the tooth.

2-…………………..

3-…………………..

4-Deeper groove.

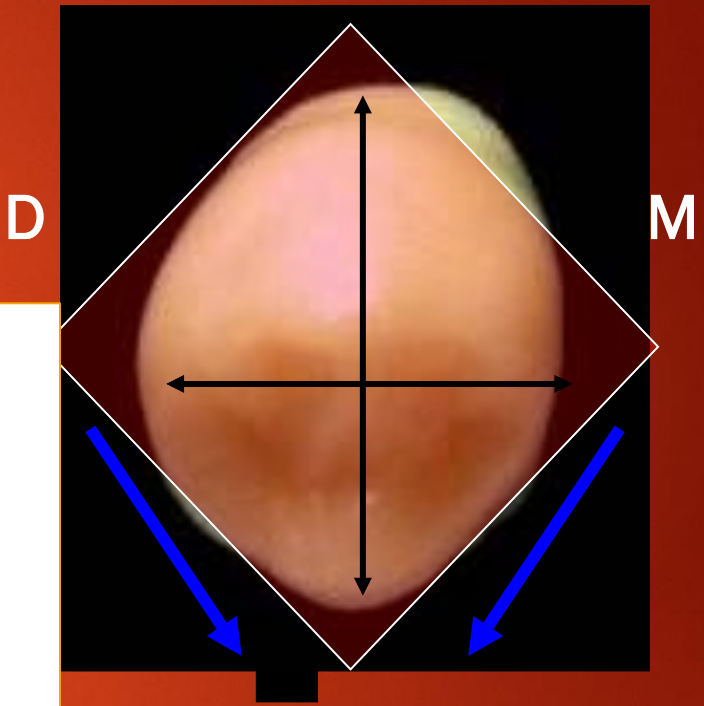


Occlusal Surface

Geometric outline

Diamond-shaped

Buccal surface is much larger than Lingual surface ***(due to sharp and severe lingual convergence)***



Crown thickness is wider than width

Surface anatomy of occlusal aspect

Elevations:

- Buccal triangular ridge
- Lingual triangular ridge
- Transverse ridge
- Mesial and distal marginal ridges

Depressions:

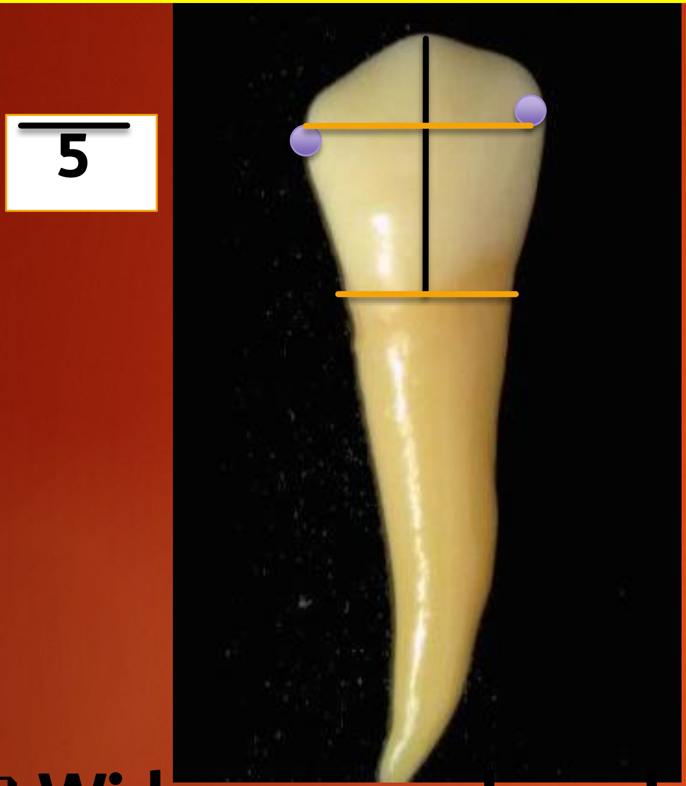
- Mesial and Distal <u>rounded</u> fossae (snake eyes).
- Two developmental pits.
- Mesiolingual developmental groove (originates from……)

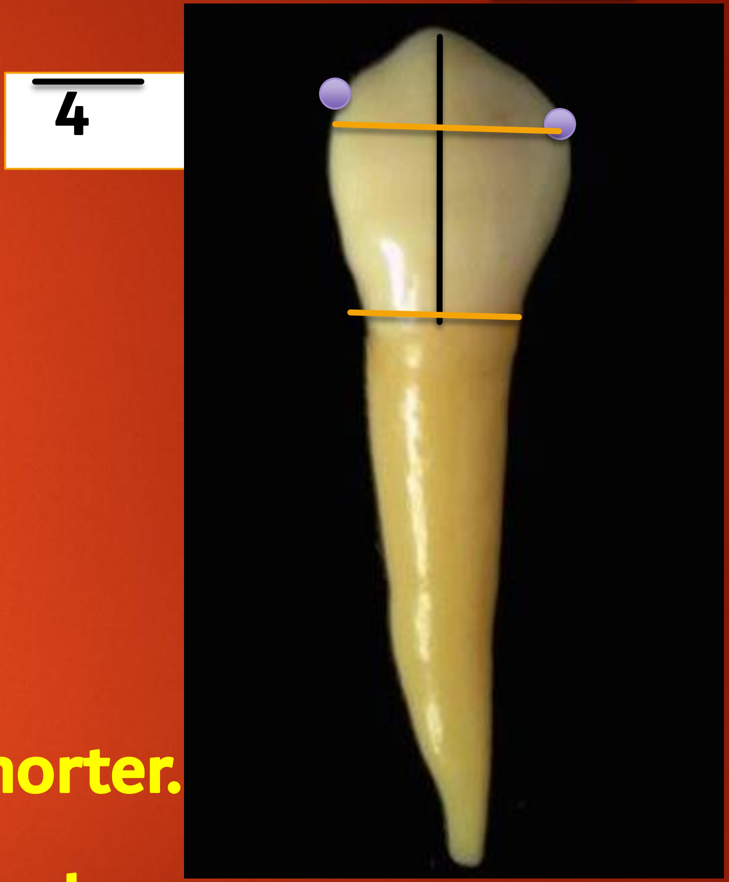
Mandibular Second Premolar

Geometric outline of Facial and lingual aspects

Trapezoid shape with the shortest uneven side cervically.

Buccal Surface





- Wider crown , broader neck but shorter.
- Buccal cusp shorter and less pointed.
- Mesial & Distal contact area more occlusally located.
- Buccal ridge less developed.

Lingual Surface

5 (with a bar over it)

Either one or two lingual cusp

Lingual cusp is nearly equal to Buccal cusp (longer than of 4)

4 (with a bar over it)

Slight lingual convergence.

The surface is convex with max. convexity at Occlusal 1/3.

Lingual Surface

5

4

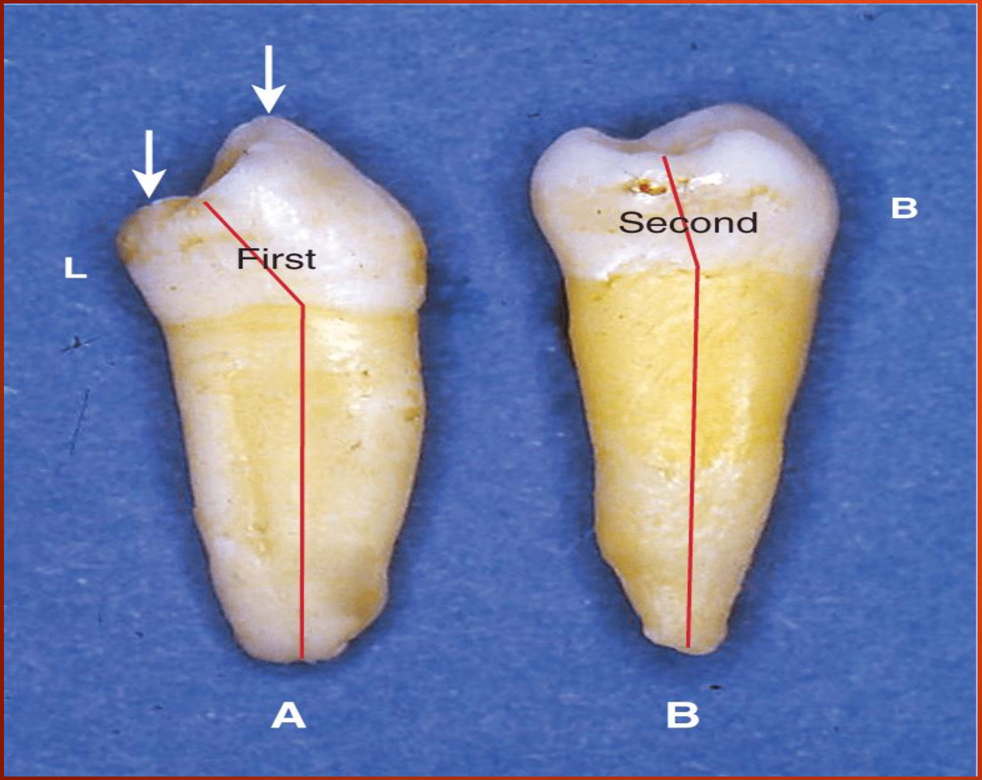
If two lingual cusp, they are named Mesiolingual and Distolingual cusps.

ML cusp is larger than DL cusp

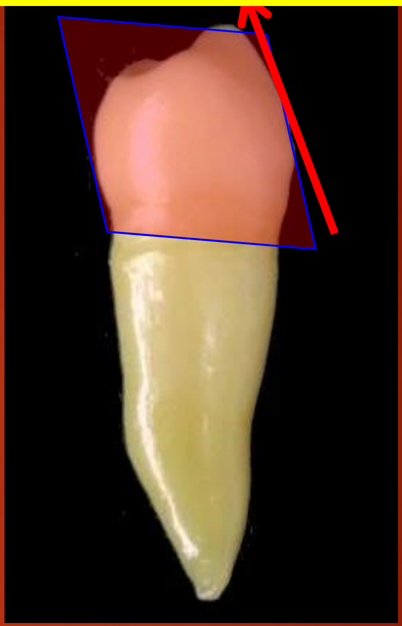
They are separated by Lingual DG

Proximal Surfaces

In comparison to lower 4



$\overline{5}$



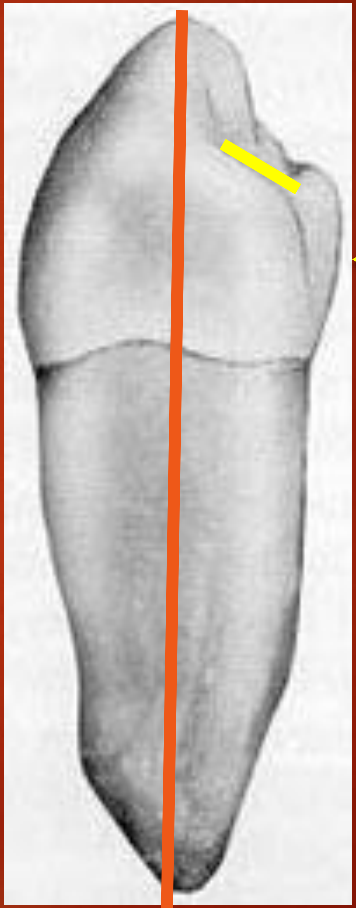
❖Geometric outline is Rhomboid

❖Less prominent lingual inclination

Mesial Surface

5

4



Lingual Maximum convexity at Occ. 1/3

ML cusp equal or slightly longer than B cusp but larger than of 4

The mesial marginal ridge straight

Buccal cusp (less pointed) tip on the line with junction of buccal & middle 1/3 of the root.

Distal Surface

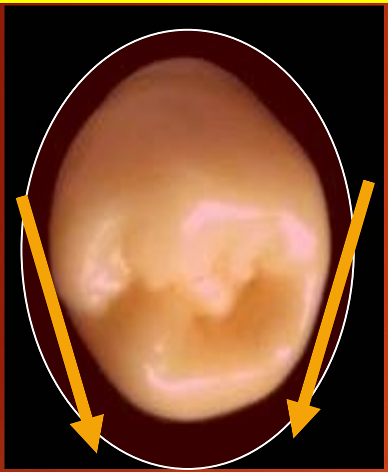
Similar to Mesial aspect except:

DL cusp is shorter and smaller than ML cusp.

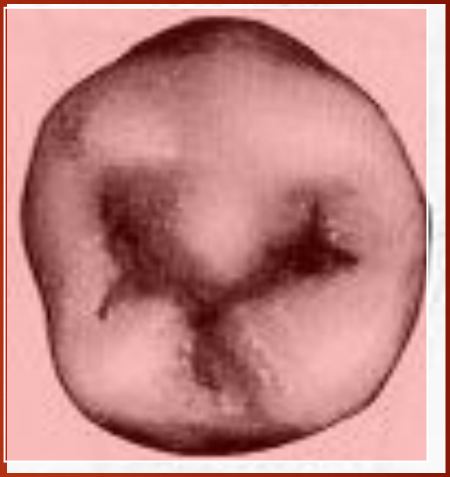
Mesial and Distal surfaces in 2 cusp type are similar to 3 cusp type (except 1 cusp seen lingually)

Surface anatomy is similar to that of $\overline{4}$ except absence of mesiolingual DG.

Occlusal Surface



Geometric outline



Two cusp type

Three cusp type

Round shaped

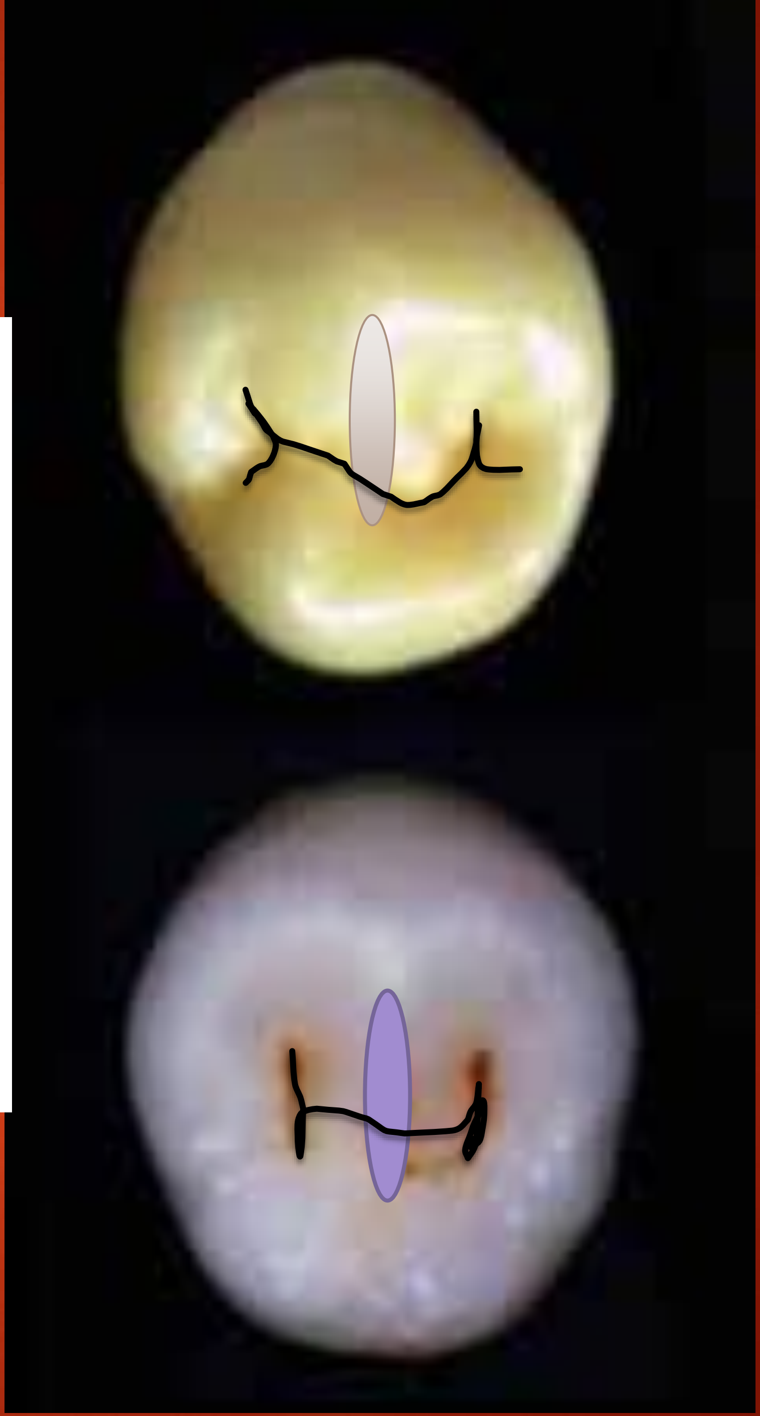
Square shaped

Slight lingual convergence

Absence lingual convergence.

Two cusp type

- Buccal cusp triangular ridge is larger than the lingual cusp ridge.
- Transverse ridge runs in the middle joining the two triangular ridges.
- The central groove is U or H shape grooves pattern.



Three cusp type

Elevations:

-Buccal triangular ridge. *(largest)*

-Lingual triangular ridges (ML &DL).

(Which is larger???)

Well developed M & D marginal ridges

Depressions:

-Developmental grooves Y shape (MB, DB & lingual developmental groove)

-Central fossa

-M and D triangular fossae

D

M

Pulp cavity of Lower Premolars

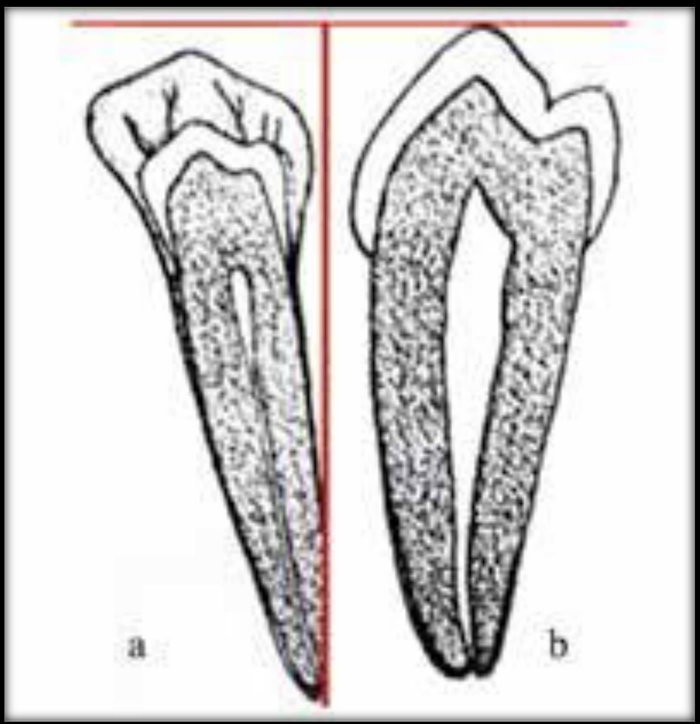
Mesio-distally:

- The pulp chamber is pointed occlusally with one pulp horn buccally and one pulp horn lingually (two pulp horns lingually in 3 cusp type lower 5).

-The root canal tapers to the apex.

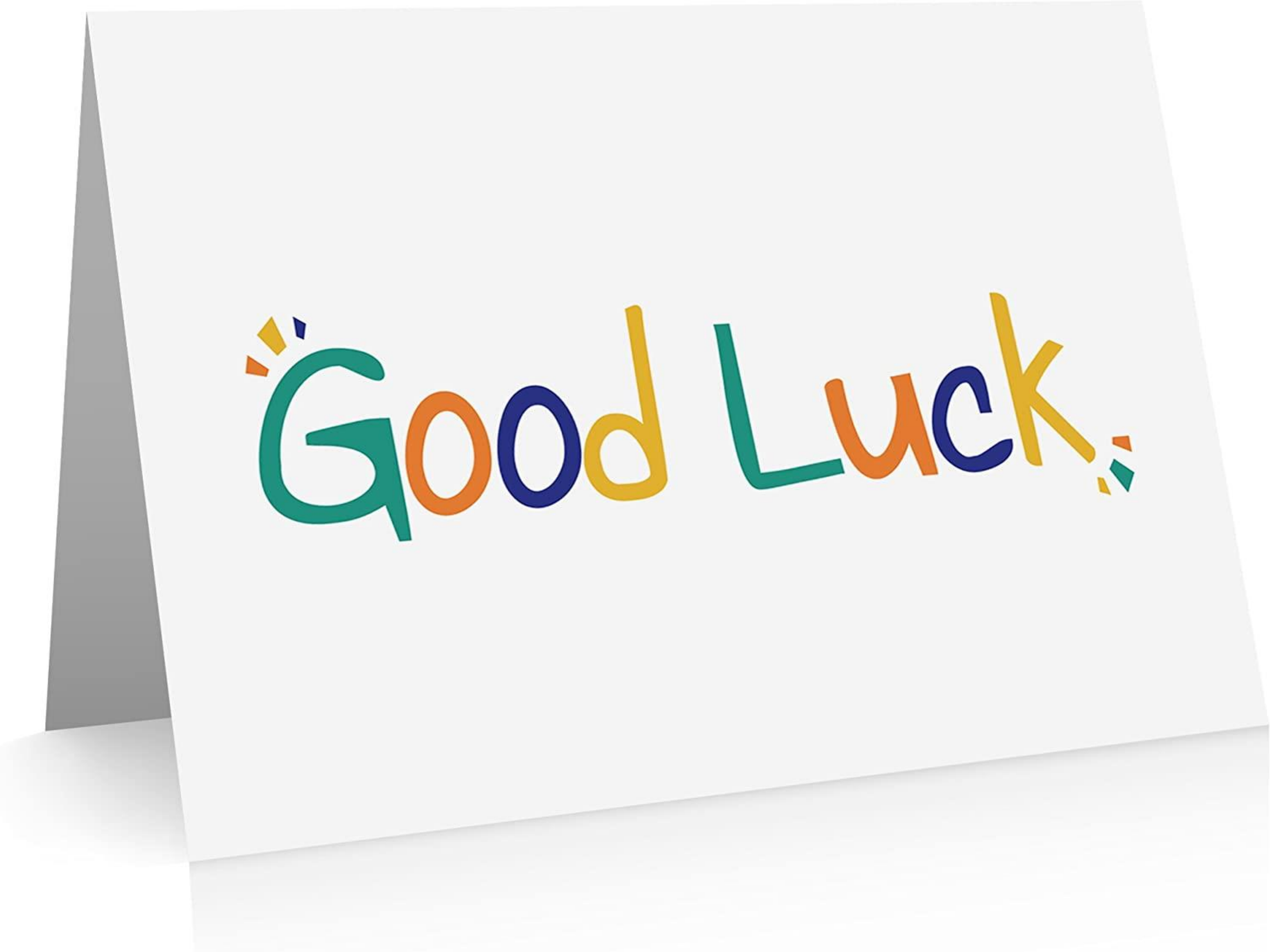
Bucco-lingually:

- The pulp chamber has two pulp horns, buccal & mesiolingual or distolingual
- The root canal tapers to the apex.



Chronology of Premolars

	Beginning of calcification	Crown completion	Eruption	Root Completion
$\frac{4}{4}$	1½- 1¾ y. / 1¾- 2 y.	3-4 y. before eruption date	$\frac{10}{10}$	3-4 y. after eruption date
$\frac{5}{5}$	2-2¼ y. / 2¼-2½ y.		$\frac{11-12}{11-12}$	


Good Luck